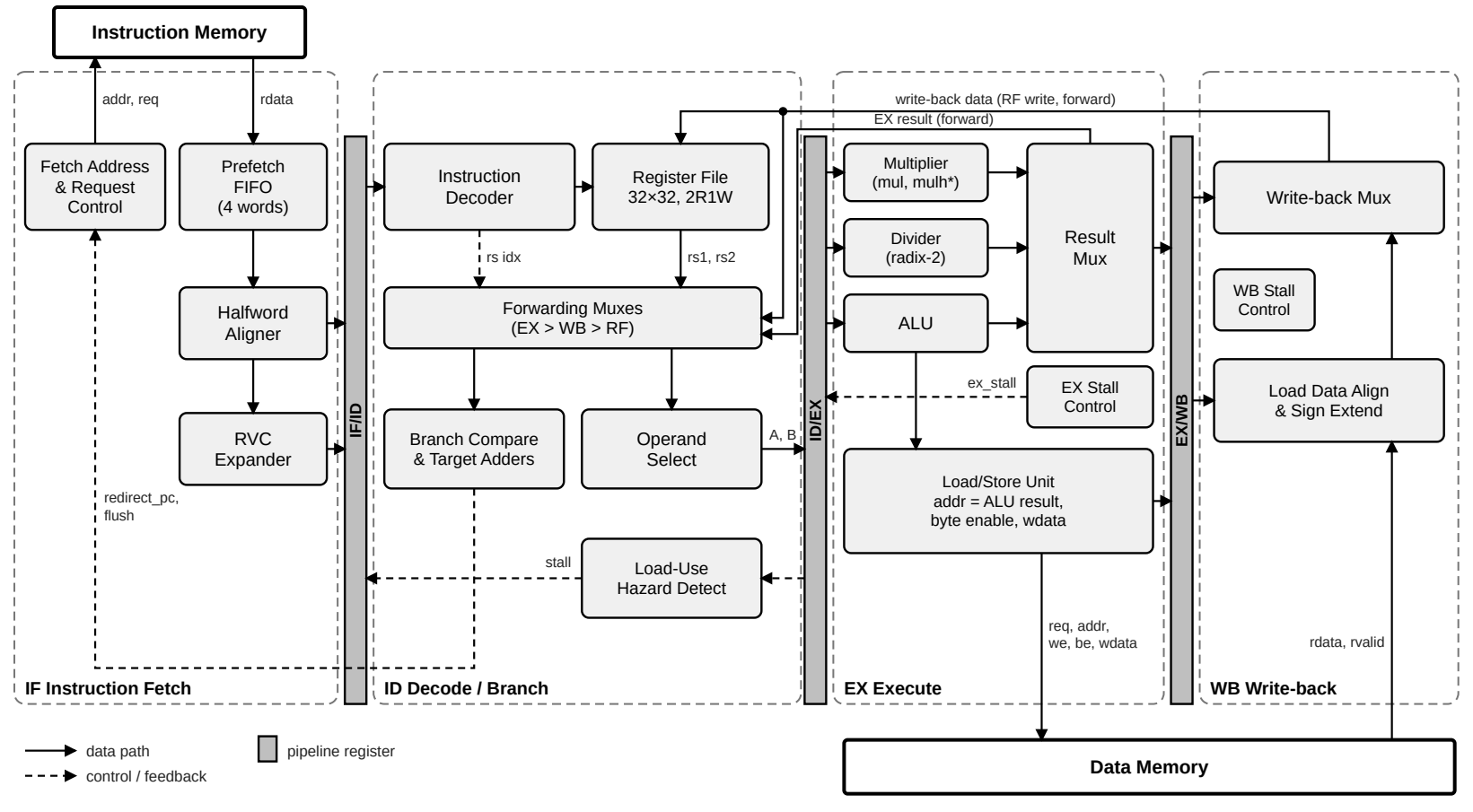
Instruction Memory
addr, req
rdata
Fetch Address & Request Control
Prefetch FIFO (4 words)
Halfword Aligner
RVC Expander
redirect_pc, flush
IF/ID
IF Instruction Fetch
Instruction Decoder
Register File 32×32, 2R1W
rs idx
rs1, rs2
Forwarding Muxes (EX > WB > RF)
Branch Compare & Target Adders
Operand Select
A, B
stall
Load-Use Hazard Detect
ID Decode / Branch
ID/EX
write-back data (RF write, forward)
EX result (forward)
Multiplier (mul, mulh*)
Divider (radix-2)
ALU
Result Mux
ex_stall
EX Stall Control
Load/Store Unit addr = ALU result, byte enable, wdata
req, addr, we, be, wdata
EX Execute
EX/WB
Write-back Mux
WB Stall Control
Load Data Align & Sign Extend
rdata, rvalid
WB Write-back
Data Memory
data path
control / feedback
pipeline register